SEC P vs NP log 2026-05-17

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-17 04:32 UTC

SEC P vs NP — daily log 2026-05-17

11 cycles recorded

GCT Lie-Stabilizer Codimension Linearly Bounds ACC^0 Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric Complexity Theory: Lie-algebra stabilizers of multilinear polynomial extensions in the Mulmuley-Sohoni-Bürgisser-Ikenmeyer framework × $ACC^0[m]$ circuit size for Boolean functions (Sipser-like targets)
- **Recorded**: 2026-05-17 00:01 UTC
- **Entry ID**: `83d26c2ef069`

Statement

For a Boolean function f:{-1,+1}^n -> {-1,+1}, let F_f(x) = Σ_{S⊆[n]} f̂(S) · Π_{i∈S} x_i be its multilinear Walsh extension over Q, let L(F_f) := {A ∈ gl_n(Q) : Σ_{i,j} A_{ij} x_j ∂_i F_f ≡ 0 in Q[x_1,…,x_n]} be its GCT Lie stabilizer, and set γ(f) := n^2 − dim_Q L(F_f). Conjecture: there is an absolute constant C ≤ 10 such that every $ACC^0[m]$ circuit of size s computing f satisfies γ(f) ≥ n^2 − C·s. A single $ACC^0[m]$ circuit of size s computing some f with γ(f) < n^2 − 10·s falsifies the conjecture.

Rationale

GCT's symmetry-characterization principle says polynomials whose stabilizer Lie algebras exceed generic dimension are atypical; we invert this to claim small ACC^0 circuits cannot accidentally synthesize multilinear extensions of anomalously high symmetry. The Lie stabilizer is GL_n-equivariant rather than truth-table-dense, so it dodges Razborov–Rudich's 'large + constructive' clause, and it is computed from the GL_n action on polynomials, not from black-box algebraic extensions — placing the invariant outside the algebrization and relativization regimes that GCT was designed to escape.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [1212.5380v2] On properties of principal elements of Frobenius Lie algebras - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [2411.11095v3] Invariant theory and coefficient algebras of Lie algebras - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1504.04131v2] Formulas for the Walsh coefficients of smooth functions and their application to bounds on the Walsh coefficients

Empirical Test

- exit code: 1, elapsed: 0.11s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 102, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 86, in run_trial
    gamma = gaussian_elimination(system)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 65, in gaussian_elim
    if matrix[i][i] == 0:
       ~~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in gaussian_elimination before any of the 480 trials or reference-function checks produced slack data, so the pre-registered support condition cannot be evaluated. Per Rule 2, a crash yields INCONCLUSIVE. | next: Fix the Gaussian elimination routine (guard against out-of-range pivot indexing and handle rectangular/short matrices) and rerun the full (n,s)-grid plus reference-function suite to obtain actual slack statistics.

2-adic Newton Slopes of DISJ Sub-blocks Scale as √n

- **Verdict**: INCONCLUSIVE
- **Bridge**: p-adic analysis (Newton polygons over Q_2 of integer characteristic polynomials) × Randomized communication complexity of DISJOINTNESS via average-case tensor invariants on sub-blocks
- **Recorded**: 2026-05-17 00:26 UTC
- **Entry ID**: efd166108501

Statement

Let M_n ∈ {-1,+1}$^{\{2}$n × 2^n} be the signed disjointness matrix M_n[x,y] = (-1)^{[x∩y≠▯]} where x,y are interpreted as subsets of [n]. For each n, draw uniformly random row set R and column set C with |R|=|C|=k=min(2n,40), form A := M_n[R,C], compute its exact integer characteristic polynomial p(t)=Σ_{i=0}^{k} c_i t^i, and define the 2-adic Newton slope count ν_2(A) := number of distinct slopes in the lower convex hull of {(i, v_2(c_i)) : c_i ≠ 0} where v_2 is the 2-adic valuation (v_2(0):=+∞). Conjecture: E_{R,C}[ν_2(M_n^{R,C})] = Θ(√n); moreover for every n ≥ 8 the DISJ median exceeds the median of ν_2 for an i.i.d. ±1 matrix of the same shape by a factor ≥ 1.5, and a single (n, seed) pair where this median ratio inverts falsifies the conjecture.

Rationale

The subset semilattice underlying DISJ injects 2-adic congruence structure into every sub-block (parity of intersection size acts like a Z_2-valued character), so subdeterminants accumulate distinguished 2-adic valuations that random Bernoulli matrices lack — a property that should propagate, via Yao's principle, into a lower bound on randomized communication. Newton polygons over Q_2 are not preserved by polynomial extension to F_2[t]/⟨t^2-t⟩, evading the algebrization barrier, and the invariant is structural rather than 'large+useful on truth tables', evading Razborov-Rudich. The bridge between p-adic Newton-polygon geometry and communication complexity has essentially no prior literature, while the average-to-worst-case framing (E[ν_2] ↦ worst-case CC lower bound) follows the Impagliazzo-Wigderson template.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 34.76s

```
tances_tested': 5, 'conjecture_holds': False, 'counterexample': 'Ratio 1.0 < 1.5 for seed 547'}
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <module>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <genexpr>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: Multiple trials report conjecture_holds=False with ratio 1.0 < 1.5 (e.g., seeds 547, 593, 631, …), satisfying the pre-registered falsification condi | next: Re-run with corrected seed-logging, stratify by n ∈ {10,14,18,22}, and check whether the ν_2 statistic saturates at small k=min(2n,40) — perhaps a normalized slope-density (ν_2/k) would better capture any DISJ-vs-random gap.

Cyclic Burnside Orbit Count Lower-Bounds AC^0 Size for PARITY

- **Verdict**: INCONCLUSIVE
- **Bridge**: Burnside-ring / equivariant orbit counting of cyclic group actions on labeled circuit DAGs (a corner of combinatorial representation theory rarely applied to circuit complexity) × AC^0 circuit size for PARITY_n, with the lifting-style intuition that gadget symmetry constrains the lifted communication matrix
- **Recorded**: 2026-05-17 00:49 UTC
- **Entry ID**: 80d12cf7a27e

Statement

Let C be an AC^0 circuit over inputs x_0,…,x_{n-1} computing PARITY_n with n a prime ≥ 3, depth(C)=d and size(C)=s, and let σ be the cyclic shift σ(i)=i+1 mod n acting on each gate g of C by simultaneously relabeling every input wire x_i appearing in g's recursive expansion to x_{σ(i)}. Define ψ(C) as the number of ⟨σ⟩-orbits of gates of C under the equivalence g ∼ h iff some σ^k maps the lex-min canonical (op-type, sorted-children) form of g to that of h. Then for every such C computing PARITY_n we conjecture ψ(C) ≥ (1/(4d)) · log_2(s), refuted by any single PARITY-computing AC^0 circuit whose ratio 4·d·ψ(C)/log_2(s) drops below 1.

Rationale

Lifting theorems translate query lower bounds into communication lower bounds by composing the function with a symmetry-breaking gadget; dually, when the target function (PARITY) is itself shift-invariant, a small circuit must reuse few genuinely distinct sub-computations, so the Z_n-orbit count of its gates becomes a Burnside-style proxy for the work that random restrictions are usually invoked to expose. Because ψ is defined on the circuit DAG rather than on the truth table, the invariant sits outside the Razborov–Rudich natural-proofs barrier; because the construction is purely combinatorial (no polynomial extension of the Boolean ring) it does not algebrize, matching the LIFTING approach's safe-direction profile.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2304.02770v2] Tight Correlation Bounds for Circuits Between AC0 and TC0 - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [1708.00951v2] Canonical Heights and Monomial Maps: On Effective Lower Bounds for Points with Dense Orbit - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2005.06373v2] Counting Schur Rings over Cyclic Groups of Semi-prime Order

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 225, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 180, in run_trial
    psi = compute_psi(circuit, n)
          ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 139, in compute_psi
    gate_hashes[gate] = hash_gate(gate)
                        ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 126, in hash_gate
    return hash(gate)
           ^^^^^^^^^^
TypeError: unhashable type: 'list'
```

Judge reasoning

The test crashed with a TypeError (unhashable list) in hash_gate before any trial completed, so no r values were produced and the pre-registered support condition could not be evaluated. | next: Fix the canonicalization in hash_gate by converting children lists to tuples (or using a frozen canonical form) before hashing, then re-run the full 2160-circuit grid.

Cauchy Mean Width of Minterm Hull Lower-Bounds Monotone k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Convex geometry (Cauchy mean-width / intrinsic 1-volume of polytopes in R^N, in the spirit of Schneider's Brunn–Minkowski theory, rarely connected to circuit complexity) × Karchmer–Wigderson / monotone DNF size and formula complexity for the k-CLIQUE indicator
- **Recorded**: 2026-05-17 01:28 UTC
- **Entry ID**: `65d1ffdf3cc6`

Statement

For a monotone Boolean function f on N variables, let $M(f) \subset \{0,1\}^N$ be its set of minterms (minimal satisfying assignments viewed as 0/1 vectors) and let $K(f) := \mathrm{conv}(M(f) \cup \{0\}) \subseteq [0,1]^N$. Define $\mu(f) := MW(K(f))^2$, where $MW(K) = 2\cdot E_{u\sim \mathrm{Unif}(S^{N-1})}[\max_{x\in K}\langle u,x\rangle]$ is the Cauchy mean width. We conjecture: (i) for every monotone DNF representation of f with s terms, $\mu(f) \le C_1\cdot\log(s+1)\cdot(\log N+1)$; (ii) for the k-CLIQUE indicator on K_v with $k=\lceil\log_2 v\rceil$ (so $N = v(v-1)/2$), $\mu(f) \ge C_2\cdot v$, for absolute constants $C_1,C_2>0$. A single instance with $\mu(f) > C_1\cdot\log(s+1)(\log N+1)$, or a k-CLIQUE indicator with $\mu < C_2\cdot v$ at $v \le 9$, falsifies the conjecture.

Rationale

Monotone DNFs are unions of axis-aligned subcubes, so conv(M(f)) is the convex hull of at most s lattice vectors of bounded coordinate sum, and its mean width should grow only logarithmically with the number of generators (a Sudakov/Talagrand-style upper bound for sparse 0/1 hulls). The k-CLIQUE minterm cloud, in contrast, lives on the (k choose 2)-th slice and is rotationally rich because v-vertex automorphisms act transitively on edges, forcing the supporting hyperplanes to spread evenly over many directions and inflating mean width linearly in v. If both bounds hold, any monotone DNF for k-CLIQUE has $s \ge 2^{\Omega(v/\log v)}$, a Razborov-flavor separation via a purely metric-geometric witness that bypasses approximator combinatorics.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `33.94s`


```
me": "mean_width_squared", "metric_value": 1.136003662687513, "instances_tested": 17, "conjecture_ho
CLIQUE with v=4 has mu=1.136003662687513 < 0.5*v=2.0", "seed": 421}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1857866298955364, "instances_tested"
CLIQUE with v=4 has mu=1.1857866298955364 < 0.5*v=2.0", "seed": 463}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1927318541604457, "instances_tested"
CLIQUE with v=4 has mu=1.1927318541604457 < 0.5*v=2.0", "seed": 503}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1396899464107255, "instances_tested"
CLIQUE with v=4 has mu=1.1396899464107255 < 0.5*v=2.0", "seed": 547}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1832069622303316, "instances_tested"
CLIQUE with v=4 has mu=1.1832069622303316 < 0.5*v=2.0", "seed": 593}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1742179582193337, "instances_tested"
CLIQUE with v=4 has mu=1.1742179582193337 < 0.5*v=2.0", "seed": 631}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1635078759135067, "instances_tested"
CLIQUE with v=4 has mu=1.1635078759135067 < 0.5*v=2.0", "seed": 677}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1855075477194135, "instances_tested"
CLIQUE with v=4 has mu=1.1855075477194135 < 0.5*v=2.0", "seed": 727}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.172332125064841, "instances_tested":
CLIQUE with v=4 has mu=1.172332125064841 < 0.5*v=2.0", "seed": 773}
TRIAL: {"metric_name": "
```


Judge reasoning

Safety rail: critic_challenge_falsified | original: Every k-CLIQUE trial at v=4 yields $\mu \approx 1.17$, well below the falsification threshold $0.25\cdot v = 1.0$... actually $0.25\cdot 4=1.0$ so this passes the $0.25\cdot v$ floo | next: Rescale the lower bound to $\mu(f) \ge C_2\cdot v/\log v$ or test the conjecture with normalized mean width $MW(K(f))/\sqrt{N}$ to remove metric-saturation artifacts at small v.

Costas Displacement Coincidence Lower-Bounds AC^0 Size for PARITY

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Costas array / Welch–Costas displacement combinatorics (Golomb–Costas radar arrays and B_2/Sidon-deficiency theory of integer sets) — a corner of additive combinatorics developed for radar/sonar that has essentially no presence in circuit complexity literature × AC^0 depth-d circuit size for PARITY_n (Hastad-style lower bounds, viewed through the lens of bottom-gate support designs)
- **Recorded**: 2026-05-17 02:01 UTC
- **Entry ID**: `6bb89da9ec2e`

Statement

Let C be a depth-d AC^0 circuit on n inputs with bottom-level (depth-1) AND/OR gates having literal-supports S_1,…,S_m ⊆ [n]. Define the Costas displacement defect κ(C) = log_2(1 + max_{δ ∈ {1,…,n-1}} Σ_{i=1}^{m} |{(a,b) ∈ S_i × S_i : a > b and a−b ≡ δ (mod n)}|). We conjecture that for every C that computes PARITY_n exactly, κ(C) ≥ (1/4) · n^{1/(d−1)}, and consequently size(C) ≥ 2^{Ω(κ(C))}. A single depth-d AC^0 circuit that computes PARITY_n while having κ(C) < (1/4) · n^{1/(d−1)} refutes the conjecture.

Rationale

PARITY is translation-invariant on the Boolean cube, so any AC^0 circuit computing it must spread its bottom-level supports densely across all cyclic displacements — exactly the property that Costas/Welch arrays minimize. Replacing Hastad's random restriction by a Costas-style design-theoretic lower bound turns AC^0 PARITY bounds into a question about Sidon-deficiency of support multisets. The invariant depends only on the circuit's bottom-support combinatorics (not the function's truth table), so it sidesteps Natural Proofs, and uses no ring operations, sidestepping algebrization.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.18s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode 124) without producing a RESULT line, so neither the support nor falsification condition could be evaluated. | next: Reduce the parameter sweep (e.g., cap n at 20 and d=2) or parallelize per-seed evaluation to produce data within the timeout.

Vertex Star-Discrepancy Equals Spectral Norm for XOR Comm

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric discrepancy theory (vertex-restricted L∞ star discrepancy of {0,1}^n point sets, in the Chazelle–Matousek tradition of axis-parallel box discrepancy) × Randomized two-party communication complexity of XOR-functions f(x,y)=g(x⊕y) accessed through the discrepancy-method lower bound disc(M_f)=||ĝ||_∞
- **Recorded**: 2026-05-17 02:29 UTC
- **Entry ID**: `d27e7e45b29b`

Statement

For any non-constant g:{0,1}^n→{−1,+1}, let P_g={z∈{0,1}^n : g(z)=−1} and define the lattice star discrepancy D*lat(g) := max{a∈{0,1}^n} | |P_g ∩ [0,a]⊓|/2^n − (|P_g|/2^{n)·(2}{popcount(a)}/2^n) |, where [0,a]⊓={z∈{0,1}^n : z≤a coordinatewise}; let ||ĝ||∞ := max{S⊆[n]} |ĝ(S)| be the spectral norm. Conjecture: there exist absolute constants c_1=1/8 and c_2=8 such that for every n≥4 and every non-constant g, c_1·||ĝ||_∞ ≤ D*lat(g) ≤ c_2·||ĝ||∞·log_2(n+1). A single (n,g) pair violating either inequality refutes the conjecture.

Rationale

The discrepancy lower bound on randomized XOR-communication is exactly log(1/||ĝ||∞), yet ||ĝ||∞ is a global Fourier quantity with no geometric face. Vertex star discrepancy is the canonical Chazelle–Matousek irregularity measure of a point set inside [0,1]^n; restricting test boxes to lattice corners makes it computable in O(n·2^n) via the zeta transform. If D*lat ≈ ||ĝ||∞, this gives the first concrete geometric (axis-parallel box) certificate of XOR-communication hardness, opening a Banaszczyk/Kashin-style geometric attack on unsaturated R(DISJ)-like bounds.

Novelty

- Judge: `NOVEL` over 2 arXiv hits

Top hits consulted: - [s2:1704.02537] Dual polynomials and communication complexity of XOR functions - [s2:19311e592db3bd54ad8685ae0d27b815c87be] Dual polynomials and communication complexity of XOR functions

Empirical Test

- exit code: `124`, elapsed: `240.03s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode=124) without producing a RESULT line or any trial data, so neither the support nor falsification condition can be evaluated. | next: Reduce the maximum n or restrict the enumeration over a∈{0,1}^n (e.g., sample or use a smarter combinatorial bound) so the 210 trials complete within the time budget.

Operator-SoS 4-Trace Gap Lower-Bounds Blocks-Order Read-Twice BP for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Sum-of-Squares / Lasserre hierarchy on operator-valued layer-product tensor moments (Putinar–Helton-style non-commutative SoS pseudo-trace invariants applied to layer-difference matrices of branching programs — a corner combining moment-SoS literature with BP structural invariants, with <5 papers on the BP side) × Read-twice oblivious branching program size for IP_2 (the open BP_READTWICE separation of Borodin–Razborov–Smolensky 1993)
- **Recorded**: 2026-05-17 02:59 UTC
- **Entry ID**: `0ecd5f06ea81`

Statement

Let P be a read-twice oblivious BP on 2n Boolean variables (x_1,…,x_n,y_1,…,y_n) with width w and length 4n, layer transition matrices M_j(0),M_j(1)∈{0,1}^{w×w}, and layer-difference matrices

N_j := M_j(1)−M_j(0); for each variable v with its two read-layers (a_v,b_v) form the paired operator P_v := N_{a_v}N_{b_v}∈R^{w×w}, and define the operator-SoS 4-trace gap ρ(P) := log_2 max_{u≠v} |⟨P_u, P_v⟩_F| / w^2, where ⟨A,B⟩_F = tr(A B^T). CONJECTURE: (Upper anchor, trivial via SoS-Cauchy-Schwarz) ρ(P) ≤ 2·log_2 width(P) + 2 for every read-twice oblivious BP P; (Lower bound, non-trivial) for every BLOCKS-ORDERED read-twice oblivious BP P (all x-layers preceding all y-layers in each of the two reads) that computes IP_2(x,y)=⊕_i x_i y_i with bias ≥ 1/2 + n^{−2}, ρ(P) ≥ n/16 − 4·log_2 n. Equivalently, any such blocks-ordered read-twice BP for IP_2 has width ≥ 2^{n/40} for all sufficiently large n; a single small-width blocks-ordered read-twice BP for IP_2 with ρ(P) < n/16 − 4·log_2 n falsifies the conjecture.

Rationale

The paired operator P_v = N_{a_v} N_{b_v} packages the two reads of variable v into a single non-commutative SoS-style moment, and the Frobenius pairing ⟨P_u,P_v⟩_F is exactly the kind of trace identity that survives Lasserre relaxation at level 2 yet remains computable directly from the BP. For IP_2 the algebraic obstruction that forces exponential read-once size — namely that x and y blocks cannot be summarized in fewer than 2^n states without bilinear cancellation — should reappear in the read-twice setting as a forced spread of mass across many variable-pairs in the Gram matrix of {P_v}, producing a heavy off-diagonal entry. Because ρ depends on the specific BP layer matrices (not the truth table) it cannot collapse to a function-symmetric invariant, side-stepping the natural-proofs trap that killed the prior Lasserre-rank attempt in this approach.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8f1e5764.py", line 113, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8f1e5764.py", line 69, in run_trial
    N = generate_bp(w, n, seed)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8f1e5764.py", line 44, in generate_bp
    N = [matrix_subtract(M[j], M[j-1]) if j > 0 else M[j] for j in range(2*n)]
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8f1e5764.py", line 28, in matrix_subtra
    w = len(A)
        ^^^^^^
TypeError: object of type 'int' has no len()
```

Judge reasoning

The test crashed with a TypeError in matrix_subtract before producing any data, so neither the upper-anchor nor the IP_2 lower-bound criterion was evaluated. No RESULT line was emitted and the pre-registered support condition cannot be assessed. | next: Fix the matrix representation bug in generate_bp/matrix_subtract (ensure M[j] is a 2D list/array of shape w×w, not an int) and re-run the 540 random instances plus the 4 explicit IP_2 BPs.

Cross-Side Fourier Symmetric-Difference Mass Lower-Bounds DISJ CC_R

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Fourier analysis of Boolean functions (Bonami–Beckner / KKL-style level-weighting on the cube, with bipartite cross-coordinate noise operators rarely used in CC) × Randomized two-party communication complexity of DISJOINTNESS (CC_R) and other partial functions on N=2^n × 2^n matrices
- **Recorded**: 2026-05-17 03:06 UTC
- **Entry ID**: `fba37ecf9a43`

Statement

For a Boolean matrix M ∈ {0,1}$^{\{2}$n × 2^n} viewed as a function f_M : {0,1}^{2n} → {0,1} with first n bits the x-input and last n bits the y-input, and for T ⊆ [2n] let T_x = T ∩ [n] and T_y = {i ∈ [n] : i+n ∈ T}, define δ(T) := |T_x △ T_y| and the cross-asymmetry invariant Ψ(f_M) := (Σ_T f_M_hat(T)^2 · 2^{δ(T)}) / (Σ_T f_M_hat(T)^2). Conjecture: there exists an absolute constant c ≥ 1/8 such that for every Boolean matrix M, CC_R^{1/3}(M) ≥ c · log_2 Ψ(f_M), and moreover Ψ(f_{DISJ_n}) = (7/6)^n exactly (so log_2 Ψ = n·log_2(7/6) ≈ 0.222 n and CC_R(DISJ_n) = Ω(n) follows). A single Boolean matrix M with a verified protocol of cost B and log_2 Ψ(f_M) > 8·B refutes the conjecture.

Rationale

Standard Fourier □_1 / spectral norm lower bounds for CC are oblivious to the bipartition structure of the input, while CC_R cares essentially about which Fourier modes 'mix' the two parties. Weighting each Fourier mass by 2^{|T_x △ T_y|} amplifies modes that genuinely couple x- and y-coordinates asymmetrically — EQ-type matrices live on the diagonal T_x = T_y and get Ψ = 1, while DISJ's product-over-coordinates structure forces a multiplicative gain of 7/6 per coordinate, giving the right Razborov-type Ω(n). The hypothesis is therefore a Fourier-analytic 'cross-side' refinement of the standard discrepancy/spectral-norm method, in the spirit of KKL-style coordinate-weighted Fourier inequalities.

Novelty

- Judge: " over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by NATURAL_PROOFS barrier

---

q-Major Cancellation Gap Separates Permanent from Determinant Supports

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Mahonian statistics on the symmetric group (major index, q-factorials, RSK cocharge) × Padded permanent/determinant comparison on shared 0/1 support matrices (GCT det-vs-perm flavor)
- **Recorded**: 2026-05-17 03:32 UTC
- **Entry ID**: `796531c3b7fa`

Statement

For an n x n matrix M in {0,1}^{n x n}, let T_M = {sigma in S_n : M[i,sigma(i)] = 1 for all i} and define the q-major permanent and signed q-major determinant on the support T_M by perm_q(M) := sum_{sigma in T_M} q^{maj(sigma)} and det_q(M) := sum_{sigma in T_M} sign(sigma) * q^{maj(sigma)}, where maj(sigma) is the major index (sum of descent positions). Conjecture: for every n >= 3 and every 0/1 matrix M with |T_M| >= 2, the q=2 cancellation ratio R_2(M) := perm_2(M) / max(1, |det_2(M)|) satisfies R_2(M) >= sqrt(|T_M|) / (2*n). Equivalently, the sign(sigma)-weighted Mahonian sum on any 0/1-support exhibits at least square-root-of-support cancellation against the unsigned version, up to a linear factor in n.

Rationale

Permanent vs determinant in GCT differs precisely by replacing sign(sigma) with 1 inside the S_n-sum; weighting both sums by q^{maj(sigma)} (a Schur-positive Hecke-algebra principal-specialization) refracts that difference through irreducible S_n-character spectra, and at q=2 one already sees [n]_2! = prod(2^k - 1) for perm_n but a much smaller signed sum for det_n. A square-root cancellation lower bound on every common support T_M would certify that determinantal coefficient sequences on permanental supports always look 'random-like' relative to their unsigned partners, giving a Mahonian witness for the per-vs-det gap that is preserved under support-preserving substitutions.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1503.00822v2] Product Ranks of the 3×3 Determinant and Permanent - [2302.11221v6] A q-analog of certain symmetric functions and one of its specializations - [1302.4419v3] The determinant of the Malliavin matrix and the determinant of the covariance matrix for multiple integrals - [0810.2731v3] Fix-Euler-Mahonian statistics on wreath products - [1101.4332v2] Mahonian Pairs

Empirical Test

- exit code: `1`, elapsed: `1.37s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 51, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in run_trial
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j]) for sigma in permutation
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in <genexpr>
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j]) for sigma in permutation
                          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in <genexpr>
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j]) for sigma in permutation
                                                                                  ^
NameError: name 'j' is not defined
```

Judge reasoning

The test crashed with a NameError ('j' undefined) in the major-index computation before any trial completed, so no data was produced to evaluate the pre-registered criterion. | next: Fix the maj(sigma) computation to correctly count descent positions (i where sigma[i] > sigma[i+1]) and rerun the 240-trial sweep.

Immanant Positivity Width Bounds Monotone KW Depth for Perfect Matching

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Stembridge–Stanley immanant theory: the λ-immanants imm_λ(S) = Σ_σ χ_λ(σ) Π_i S[i,σ(i)] organise matchings of a 0/1 matrix into Schur–Weyl S_n-isotypic cells (a corner of symmetric-group representation theory with very few papers connecting it to monotone circuit lower bounds outside of GCT speculation) × Monotone Karchmer–Wigderson tree depth d_KW(S) at a 1-input S of the bipartite perfect-matching function PM_n, equivalent by KW to log_2 of the local monotone formula leaf size restricted to matchings inside S vs. small bipartite vertex covers
- **Recorded**: 2026-05-17 04:06 UTC
- **Entry ID**: `15b45d9e4966`

Statement

For 3 ≤ n ≤ 6 and any n×n 0/1 matrix S with perm_n(S) ≥ 1, let Σ(S) := {σ ∈ S_n : S[i,σ(i)] = 1 ∀i} and define the immanant positivity width w(S) := #{λ ⊢ n : imm_λ(S) > 0}. Let d_KW(S) be the minimum depth of any KW protocol that on Alice-input τ ∈ Σ(S) and Bob-input C, an (n−1)-vertex cover of the bipartite complement of some 0-input S′, outputs an edge (i, j) ∈ τ C. Conjecture: d_KW(S) ≥ ⌈log_2 w(S)⌉ for every such S, with equality only when Σ(S) is RSK-shape-monochromatic; in particular w(I_n) = p(n), forcing d_KW(I_n) ≥ ⌈log_2 p(n)⌉ = Θ(√n), a representation-theoretic obstruction in the spirit of GCT's perm/det separation.

Rationale

Each rectangle of a KW protocol pools matchings that all hit a common edge of Bob's cover; we conjecture this pooling is incompatible with carrying more than one irreducible χ_λ-component, so the number of non-vanishing λ-immanants is a Schur-Weyl obstruction to small KW trees. This mirrors the GCT philosophy that perm_n's orbit closure is Schur-rich while det_m's is sign-isotypic: a small substitution-stable Schur-width invariant would translate directly into a perm-vs-det monotone gap. The link of immanants to KW depth is, to our knowledge, untouched in the literature.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2003.09602v2] The Number of Perfect Matchings in Möbius Ladders and Prisms - [2304.05285v1] Hook-Shape Immanant Characters from Stanley-Stembridge Characters - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70051383.py", line 178, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70051383.py", line 132, in run_trial
    d_KW = kw_depth(matrix, covers)
           ^^^^^^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70051383.py", line 104, in kw_depth
    subsets[cover].append(sigma)
    ~~~~~~~^^^^^^^
TypeError: unhashable type: 'set'
```

Judge reasoning

The test crashed with a TypeError ('unhashable type: set') in kw_depth before producing any data, so no metrics were computed and the pre-registered support condition cannot be evaluated. | next: Fix the kw_depth implementation by using a hashable key (e.g., frozenset) for cover keys in the subsets dict, then rerun the 360-matrix sweep.

Curto-Fialkow Hankel Defect Separates Read-Twice IP_2 BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Curto-Fialkow truncated moment problem theory (Hankel positivity and flat-extension rank, the real-algebraic-geometry foundation of the Lasserre SoS hierarchy; Curto-Fialkow 1996, Lasserre 2001) applied to layer-conditional empirical moments of branching programs × Read-twice oblivious branching program size for IP_2 on 2n variables in adversarial order x_1,…,x_n,y_1,…,y_n (Borodin-Razborov-Smolensky 1993 open separation)
- **Recorded**: 2026-05-17 04:32 UTC
- **Entry ID**: `9f09553bfd37`

Statement

Let P be a read-twice oblivious BP with state set Q (|Q|=s) computing a Boolean function on N=2n variables, and let q(z,□) denote its state on input z after □ steps. For each state q∈Q and each layer □∈{1,…,2N}, define the centered conditional moment sequence m^{q,□}*k(P) = 2^{-N} Σ_z □[q(z,□)=q]·(IP_2(z)-1/2)^k for k=0,…,⌊N/4⌋, and let H^{q,□}(P) be the (d+1)×(d+1) Hankel matrix with d=⌊N/8⌋ and entries m^{q,□}*{i+j}. Define the positivity defect δ(P) = max_{q,□} #{negative eigenvalues of H^{q,□}(P)}. Conjecture: (i) for every read-twice oblivious BP P of size s, δ(P) ≤ 8·⌈log_2(s+1)⌉; (ii) every read-twice oblivious BP P that exactly computes IP_2 in the adversarial schedule satisfies δ(P) ≥ ⌈√N/2⌉.

Rationale

Curto-Fialkow theory equates degree-2d SoS feasibility with the existence of flat positive extensions of truncated Hankel sequences, so a small BP forces low-rank flat extensions of its layer-conditional empirical moments — a structural constraint never previously exploited for BP lower bounds. IP_2's bilinear ⊕-structure should force every conditional sub-population reaching a given state at a given layer to retain non-trivial bilinear correlation between the unread x-bits and the moments, obstructing flat positive Hankel extension and producing many negative eigenvalues. The conjectured gap log(s) vs √N would yield s ≥ 2^{Ω(√N)} for read-twice IP_2, attacking the BRS-1993 problem via the SoS-moment side rather than via communication or partition arguments.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8925cd5b.py", line 155, in <module>
    result = run_trial(int(seed))
             ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8925cd5b.py", line 104, in run_trial
    P, Q, N = generate_random_bp(n, w, seed)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8925cd5b.py", line 64, in generate_rand
    P[(q, l)][k] += 1
    ~~~~~~~~~^^^
TypeError: list indices must be integers or slices, not float
```

Judge reasoning

The test crashed with a TypeError in generate_random_bp before producing any data, so neither the support nor falsification condition could be evaluated. | next: Fix the indexing bug in generate_random_bp (cast the index k to int) and rerun the pre-registered sweep.